



Qtec Solution Limited

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DevOps Engineer Practical Task

Objective

Design and deploy a small production style system demonstrating **containerization, CI/CD automation, traffic management, and basic observability**.

The system should expose **one GET API and one POST API** and demonstrate how the system could handle **~100 requests/sec** while supporting **zero-downtime deployment**.

Core Task (Required)

1. Build a Simple API

Create a small service with:

- **GET endpoint** (example: [/status](#))
- **POST endpoint** (example: [/data](#))

The service can be written in any language (Node.js / Python / Go / etc.).

Explain briefly in the README how the system can handle **~100 requests/sec** (scaling strategy or configuration).

2. Containerize the Application

Package the application into a container.



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Example tool: **Docker**

Requirements:

- Production-ready **Dockerfile**
- Environment variable configuration
- Application should run via container

3. Reverse Proxy Setup

Deploy the application behind a reverse proxy.

Example tool: **Nginx**

Responsibilities:

- Route traffic to the application container
- Basic load-balancing configuration

4. CI/CD Pipeline

Create an automated pipeline that:

1. Builds the application
2. Runs basic tests
3. Builds the container image
4. Deploys the updated application

Example tools:

- GitHub Actions
- GitLab CI/CD
- Jenkins



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Explain how deployment happens **without interrupting users**.

5. Basic Monitoring & Logs

Add a basic way to observe the system.

Examples:

- Application logs
- Request logs
- Basic service health monitoring

Possible tools:

- Prometheus
- Grafana
- Container logs

Cloud Deployment

Deploy the system on a cloud provider.

Example:

- Amazon Web Services
- Google Cloud Platform
- DigitalOcean

Bonus (Optional)

Container Orchestration

Deploy using a container orchestrator.



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Example: Kubernetes

Possible features:

- Deployment manifests
- Rolling updates
- Horizontal scaling

Infrastructure as Code

Provision infrastructure using IaC tools.

Examples:

- OpenTofu
- Terraform

Example use cases:

- Provision cloud instances
- Configure networking
- Automate infrastructure deployment

Secrets & Security Management

Demonstrate secure handling of secrets.

Example tools:

- HashiCorp Vault
- AWS Secrets Manager

Security practices may include:

- No hardcoded credentials



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- Secure environment variables
 - Non-root containers

Submission Instructions

Submit a **Git repository** containing:

- Application source code
- Dockerfile
- Proxy configuration
- CI/CD configuration
- Deployment setup

README should explain:

- System architecture
- Containerization approach
- Deployment process
- How zero-downtime deployment works
- Logging / monitoring setup
- How the system can handle ~100 requests/sec

Deadline: ASAP.